

COURSE 3 - DATA ENGINEERING 2026

C3-073 - ShipmentStream Protected Review

Open only after the complete Mini-Lab attempt is saved.

Artifact	C3-073-REVIEW
Status	2026 - LEARNER EDITION
Phase	C3-06 Streaming & Event-Driven Pipelines 13 lessons 34-46 focused hours
Prerequisite	C3-05 Gate PASS; C2B streaming/lakehouse concepts may be recalled but Kafka/Spark streaming implementation is taught here.
Current lane	Apache Kafka 4.3.1 Spark Structured Streaming 4.2.0 Delta Lake 4.4.0 Debezium 3.6.2.Final
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: mental picture -> tiny event example -> guided Kafka/streaming action -> expected result -> why it worked -> break/fix -> independent changed case -> validation -> replay/state/production reasoning -> evidence. A running stream is not mastery if duplicates, late data, offsets, checkpoints or sink state are wrong.

Reviewer expectations

- Source row and unique event-id controls reconcile exactly from the supplied fixture.
- Duplicate event is visible and cannot double-create shipment state.
- Event-time vs ingest-time distinction is explicit; late-event policy is explainable.
- Kafka design states key/partitions/group/retention/replay and RF=1 lab boundary.
- Structured Streaming query/checkpoint evidence is real if claimed; simulator output is labeled separately.
- Bronze/Silver checkpoints and sink idempotency boundaries are separate and reproducible.
- Runbook covers detect -> diagnose -> recover -> validate with retained evidence.

Changed recovery

If one competency fails, repair only that skill and solve a changed shipment micro-case. Completion of repair never auto-opens a fresh full Gate retest.